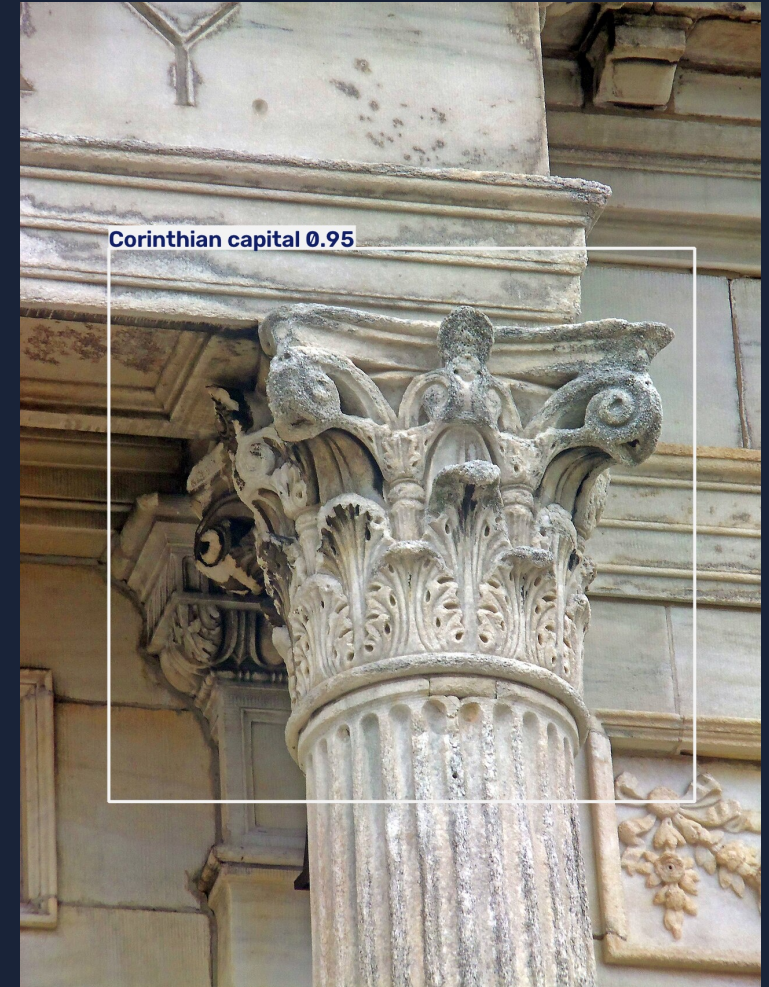


Finding classical column capitals

A small AECO visual inventory prototype

Doric | Ionic | Corinthian



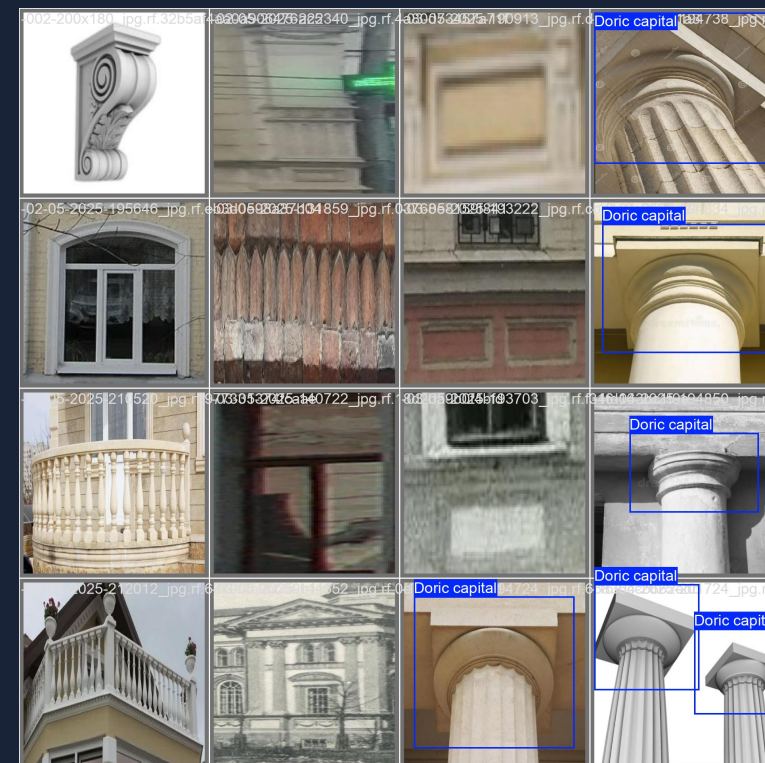
The task and the frozen data

Locate and classify capitals in architectural photos.

862 training + 216 validation images (80/20).

Source: Roboflow Architecture Segmentation v1.

Public dataset release with SHA256 verification.



Training and reproducibility

YOLOv8n, 30 epochs, batch 16, 640 px.

Original training: Tesla T4, ~8.6 minutes.

Released best.pt loads in fresh Colab without Drive or a Roboflow API key.

Full retraining remains optional in the notebook.

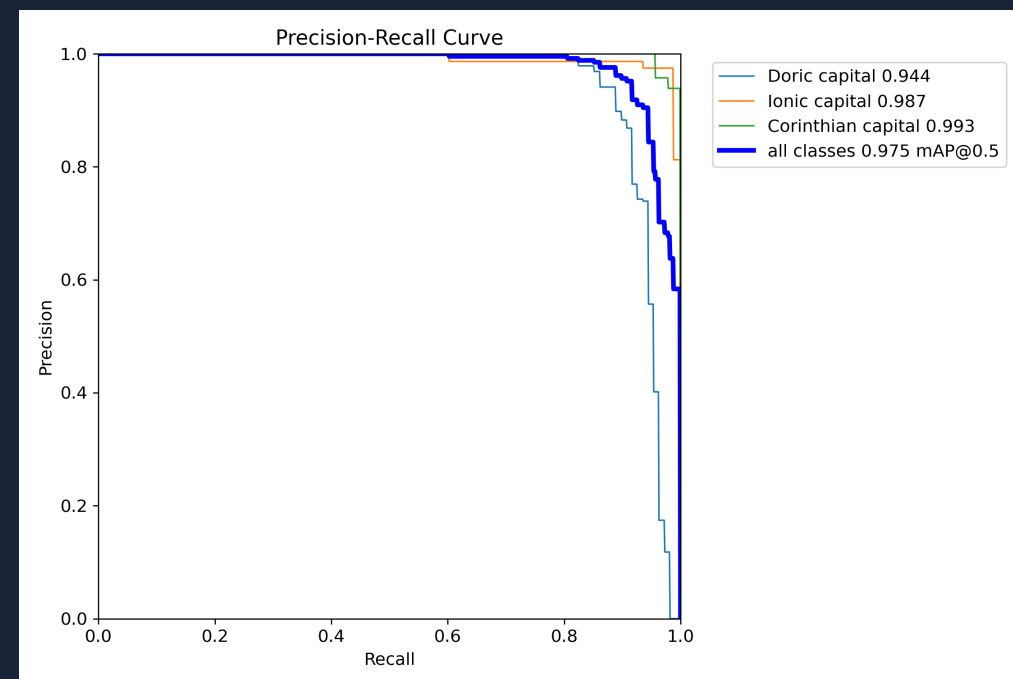
Validation: 216 images

Precision 0.960 Recall 0.939

mAP50 0.975 mAP50-95 0.694

Doric has the lowest recall: 0.859.

These numbers describe this validation split.



Five external photos expose misses

Detected: Ionic (0.44) and Corinthian (0.95).

Missed: two weathered/detached Doric capitals and one close-up Ionic capital.

Reducing confidence to 0.10 did not help.



Limits and next improvements

1. Add weathered and detached Doric examples.
2. Add Ionic close-ups and hard negatives.
3. Split evaluation by building/site; audit near duplicates.

Use only with human review; do not infer a building style or structural condition from these detections.

Source and image licenses: see docs/image_credits.md in repository.